

Development of an Intelligent Digital Dashboard Using Design Thinking to Improve Production Visibility in Automotive Aftermarket Manufacturing

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ABSTRACT

The increasing complexity of automotive aftermarket manufacturing requires companies to improve production visibility and decision-making through digital transformation. However, many manufacturers still depend on fragmented information systems, manual data recording, and limited real-time monitoring, leading to production delays, inaccurate material planning, and inefficient delivery performance. This study aims to develop an intelligent digital dashboard that improves production visibility using a user-centered approach based on the Design Thinking methodology. A qualitative case study was conducted at PT Adyawinsa Stamping Industries, an automotive component manufacturer. The Design Thinking framework, consisting of the Empathize, Define, Ideate, Prototype, Test, and Reflect stages, was applied to identify user needs, design an interactive dashboard, and evaluate its usability. Data were collected through observations, semi-structured interviews, workshops, document analysis, and usability testing involving production managers, operators, administrators, marketing staff, purchasing personnel, and information technology specialists. The developed dashboard integrates purchase order management, material availability, production progress, barcode-based work-in-process monitoring, delivery scheduling, and key performance indicator visualization into a single digital platform. The evaluation results show that the dashboard improves production visibility, information transparency, communication between departments, and production monitoring efficiency. It also supports faster and more accurate managerial decision-making by providing real-time production information. The findings demonstrate that combining Design Thinking with intelligent dashboard development offers an effective strategy for digital transformation in automotive aftermarket manufacturing. This research contributes to the development of practical, user-centered decision support systems that enhance operational performance and support Industry 4.0 implementation in manufacturing environments.

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1. Introduction

The rapid evolution of Industry 4.0 has fundamentally transformed manufacturing operations by integrating cyber-physical systems, cloud computing, artificial intelligence, big data analytics, and industrial Internet of Things (IIoT) technologies into production environments. Manufacturing organizations are increasingly expected to improve operational efficiency through intelligent information systems capable of providing real-time production visibility and supporting data-driven decision-making. Digital transformation has therefore become a strategic necessity rather than merely a technological innovation, particularly within highly competitive industries such as automotive manufacturing.

Among various manufacturing sectors, the automotive industry faces one of the most complex production environments due to extensive supply chains, strict quality requirements, fluctuating customer demand, and increasingly diverse product portfolios. Besides producing Original Equipment Manufacturer (OEM) components, automotive companies must also manage aftermarket service parts that support vehicle maintenance throughout the product lifecycle. Unlike OEM production, aftermarket production is characterized by uncertain demand, long product life cycles, numerous stock keeping units (SKUs), and strict delivery requirements. Consequently, maintaining production visibility throughout the service part manufacturing process is critical for ensuring customer satisfaction and business sustainability.

Production visibility refers to an organization's capability to continuously monitor production status, inventory availability, work-in-process conditions, resource utilization, and delivery progress using integrated digital information. Previous studies have demonstrated that improved production visibility enhances manufacturing flexibility, reduces production delays, minimizes inventory shortages, and supports proactive decision-making. However, many manufacturing organizations continue to experience fragmented production information because operational data remain distributed across multiple departments and disconnected information systems.

PT Adyawinsa Stamping Industries (PT ASI), one of Indonesia's leading automotive component manufacturers, encounters similar operational challenges in managing service part production. Although the company has implemented an Enterprise Resource Planning (ERP) system for several administrative functions, production planning, work-in-process monitoring, material tracking, and delivery scheduling are still performed manually using spreadsheet applications and informal communication channels. This fragmented information flow significantly reduces production transparency and complicates coordination among production, purchasing, logistics, marketing, and information technology departments.

The company also experiences considerable fluctuations in service part demand from Authorized Sole Distributors (Agen Tunggal Pemegang Merek—ATPM). Since demand forecasts are frequently revised and production planning relies heavily on manual data consolidation, raw material procurement often becomes inaccurate. These conditions result in excessive inventory for some materials while causing shortages of others, ultimately delaying production schedules and customer deliveries. Furthermore, limited visibility of work-in-process activities prevents managers from identifying production bottlenecks promptly, making corrective actions reactive rather than preventive.

The operational consequences of these limitations are reflected in the company's purchase order fulfillment performance, where production frequently fails to satisfy delivery targets because of insufficient coordination among departments, delayed material procurement, and the absence of real-time production monitoring. Such problems not only increase operational costs but also reduce customer satisfaction and negatively affect long-term business competitiveness. These findings indicate that technological improvements should not merely focus on automation but also emphasize integrated information visibility and user-oriented decision support.

Numerous studies have investigated Enterprise Resource Planning (ERP), Manufacturing Execution Systems (MES), Lean Manufacturing, and Digital Dashboard technologies to improve manufacturing performance. ERP systems successfully integrate organizational data and standardize business processes, whereas digital dashboards provide visual representations of operational information that facilitate managerial decision-making. Likewise, Lean Manufacturing emphasizes waste reduction through continuous process improvement, while Industry 4.0 technologies enable real-time data acquisition from production environments.

This study contributes theoretically by extending the application of Design Thinking within intelligent manufacturing system development. Unlike previous research focusing primarily on ERP implementation or dashboard visualization, this study integrates human-centered design principles into intelligent production monitoring systems.

Practically, the proposed dashboard provides manufacturing organizations with a comprehensive decision support platform that improves information transparency, cross-functional collaboration, and production visibility. The findings offer practical guidance for organizations pursuing digital transformation initiatives in automotive aftermarket manufacturing and demonstrate how user-centered dashboard development can enhance operational performance without requiring complete ERP system replacement.

2. Method

This study employed a qualitative case study approach to investigate the operational challenges encountered in automotive aftermarket production and to develop an intelligent digital dashboard that improves production visibility through a human-centered design process. A case study was considered the most appropriate research strategy because it enables an in-depth investigation of complex organizational phenomena within their real operational context. Unlike quantitative approaches that primarily examine causal relationships among predefined variables, qualitative case studies facilitate a comprehensive understanding of organizational processes, user experiences, communication patterns, and decision-making mechanisms that cannot be adequately captured through numerical data alone.

The research was conducted at PT Adyawinsa Stamping Industries (PT ASI), an Indonesian automotive component manufacturer specializing in metal stamping products for four-wheel vehicles. The company manufactures both Original Equipment Manufacturer (OEM) components and aftermarket service parts supplied to several Authorized Sole Distributors (ATPM). The aftermarket production environment presents a unique operational context because production planning is highly dependent on fluctuating customer demand, long product life cycles, extensive product variants, and strict delivery schedules. These characteristics create significant challenges in production planning, inventory management, procurement coordination, and work-in-process monitoring.

Rather than developing a completely new Enterprise Resource Planning (ERP) system, this research focused on designing an intelligent dashboard that complements the existing ERP infrastructure by providing real-time visualization of production activities and supporting managerial decision-making. Consequently, the study emphasized identifying user requirements, improving information transparency, and facilitating cross-functional coordination among departments involved in service part production.

The overall research framework adopted the Design Thinking methodology, which consists of six iterative stages: Empathize, Define, Ideate, Prototype, Test, and Reflect. This framework was selected because it prioritizes user needs throughout the system development process while encouraging iterative refinement based on continuous stakeholder feedback. The combination of qualitative case study and Design Thinking provides both methodological rigor and practical relevance, allowing the proposed dashboard to address actual operational challenges rather than merely implementing technological features.

The research was conducted within the Service Part Division of PT Adyawinsa Stamping Industries, where production activities involve multiple functional departments responsible for production planning, procurement, manufacturing operations, logistics, marketing, and information technology. Since the objective of the study was to improve production visibility across the entire production lifecycle, participants were intentionally selected using purposive sampling based on their direct involvement in service part operations and their responsibilities in production decision-making.

Six categories of stakeholders participated in this study, including the Service Part Manager, Service Part Administrator, Production Operator, Information and Communication Technology (ICT) personnel, Marketing Department representatives, and Purchasing Department representatives. Each participant possesses different operational responsibilities and interacts with the production system from distinct perspectives. Managers are responsible for production planning and overall operational control, while administrators coordinate production schedules and documentation. Production operators execute manufacturing activities and monitor production progress on the shop floor. Marketing personnel communicate customer demand and purchase orders received from ATPM,

whereas purchasing personnel manage raw material procurement and supplier coordination. ICT personnel oversee the maintenance and integration of existing information systems.

The involvement of multiple stakeholders enabled the study to capture comprehensive insights into operational problems occurring across the production process rather than focusing on a single department. Furthermore, involving cross-functional users throughout the Design Thinking process ensured that the proposed dashboard accommodates diverse information requirements and supports collaborative decision-making among departments. This multidisciplinary participation is particularly important because production visibility depends not only on technological integration but also on effective communication among organizational units.

Data collection was conducted using multiple qualitative techniques to ensure data triangulation and improve the credibility of research findings. Four primary methods were employed: direct observation, semi-structured interviews, document analysis, and participatory workshops.

Direct observations were carried out throughout the service part production process to understand existing workflows, identify operational bottlenecks, and examine how production information was generated, recorded, and communicated among departments. During these observations, researchers documented production activities from purchase order receipt through raw material procurement, manufacturing processes, packaging, warehousing, and product delivery. Particular attention was given to manual recording procedures, production tracking practices, inventory monitoring, and interdepartmental communication.

Semi-structured interviews constituted the primary source of qualitative data. Interview guidelines were developed based on the Empathize stage of Design Thinking and employed the 5W+1H questioning technique to explore users' daily activities, operational challenges, information requirements, decision-making processes, and expectations regarding future digital systems. The flexible interview structure allowed participants to describe their experiences in detail while enabling researchers to explore emerging issues through follow-up questions. These interviews generated rich qualitative insights that later informed problem definition and dashboard feature development.

Document analysis complemented observational and interview data by examining production schedules, purchase order reports, inventory records, delivery performance reports, standard operating procedures, and internal production documentation. Historical production data were also analyzed to identify recurring operational issues and evaluate organizational performance before dashboard implementation.

In addition to interviews and observations, collaborative workshops involving management and operational personnel were conducted during the Ideate phase. These workshops facilitated brainstorming sessions, priority identification, and collective evaluation of proposed dashboard functionalities. The workshops also served as an initial validation mechanism to ensure that proposed solutions aligned with organizational objectives and operational requirements.

The Qualitative data were analyzed using thematic analysis. Interview transcripts, observation notes, workshop outcomes, and organizational documents were systematically coded to identify recurring themes related to production planning, inventory management, production monitoring, interdepartmental communication, and decision-making processes. Similar themes were grouped into categories representing user requirements, operational challenges, and opportunities for system improvement. Throughout the analysis, findings obtained from different data sources were continuously compared using methodological triangulation to ensure consistency and credibility. User requirements identified during interviews were validated through observations and organizational documentation, while workshop discussions served to confirm the practical relevance of proposed dashboard functionalities.



Fig 1. Research Method

The effectiveness of the proposed intelligent dashboard was evaluated using both qualitative and operational performance indicators. Usability evaluation focused on user satisfaction, ease of use, interface clarity, information accessibility, and perceived usefulness. Feedback Capture Grid methodology enabled participants to classify dashboard features according to what worked well, what required improvement, questions that remained unanswered, and additional suggestions for future development.

Operational performance evaluation employed Key Performance Indicators (KPIs) before and after prototype implementation. The assessment focused on indicators directly related to production visibility, including purchase order fulfillment, production monitoring efficiency, work-in-process traceability, interdepartmental communication effectiveness, production scheduling responsiveness, and delivery performance. Comparing operational conditions before and after dashboard implementation allowed researchers to assess the practical contribution of the proposed system toward improving manufacturing performance.

The integration of qualitative user evaluation with KPI-based performance assessment provided comprehensive evidence regarding both the usability and organizational impact of the intelligent digital dashboard, thereby strengthening the validity of the proposed solution as a user-centered decision support system for automotive aftermarket manufacturing.

3. Results and Discussion

After The preliminary investigation revealed that the existing service part production system at PT Adyawinsa Stamping Industries was characterized by fragmented information management, manual production monitoring, and limited real-time visibility throughout the manufacturing process. Although the company had implemented an Enterprise Resource Planning (ERP) system for administrative activities, most operational information associated with production planning, work-in-process (WIP) tracking, inventory monitoring, and delivery scheduling remained disconnected across departments.

Figure 2 illustrates the existing production workflow identified during field observations. The manufacturing process begins with the receipt of purchase orders (POs) from Authorized Sole Distributors (ATPM), followed by material verification, procurement, production planning, manufacturing operations, warehousing, packaging, and final product delivery. Despite following

standardized production procedures, operational coordination heavily depended on manual documentation, spreadsheet applications, telephone communication, and messaging platforms rather than integrated digital information systems. Consequently, production information was frequently delayed, duplicated, or inconsistent among departments.

The production process involved several organizational units, including the Marketing Department, Service Part Administration, Purchasing Department, Production Division, Warehouse, and Logistics Unit. Each department maintained its own operational records independently, resulting in multiple versions of production information. Managers often required significant time to consolidate data before making production decisions, while production operators experienced difficulties identifying the current status of work-in-process components due to the absence of centralized production monitoring.

Observation results further indicated that material availability was verified manually before production scheduling. Inventory confirmation frequently required physical inspections within warehouse areas because inventory information stored in spreadsheets was not updated in real time. This practice substantially increased production planning time and occasionally resulted in incorrect procurement decisions. Similarly, monitoring production progress required supervisors to physically inspect multiple production stations rather than accessing integrated production information digitally.

Another significant limitation involved communication among departments. Purchase order revisions received from ATPM were commonly distributed through email or messaging applications, requiring administrators to manually update production schedules. Since production information was not synchronized automatically across departments, schedule changes occasionally failed to reach relevant personnel promptly, increasing the likelihood of production delays and delivery disruptions.

These findings demonstrate that operational inefficiencies were primarily associated with inadequate production visibility rather than production capacity limitations. Therefore, improving information integration and providing real-time production monitoring became essential requirements for supporting operational decision-making.

3.1 Ideate: Intelligent Dashboard Concept Development

Following problem identification, collaborative brainstorming sessions were conducted involving representatives from all participating departments. Using How-Might-We (HMW) analysis and Impact-Effort Matrix evaluation, researchers and stakeholders generated multiple solution alternatives before selecting dashboard functionalities with the highest organizational impact.

The proposed intelligent dashboard consists of six integrated modules:

1. Purchase Order Monitoring, enabling users to monitor ATPM orders, production priorities, and fulfillment status.
2. Material Availability Dashboard, providing real-time inventory information, material shortages, supplier status, and procurement progress.
3. Production Tracking Module, visualizing work-in-process activities across every manufacturing stage.
4. Barcode-Based Work-in-Process Monitoring, enabling automatic identification of product location and production status.
5. Delivery Monitoring Dashboard, displaying shipment schedules, delivery progress, and customer fulfillment performance.
6. Key Performance Indicator (KPI) Dashboard, summarizing production efficiency, purchase order fulfillment, production delays, inventory performance, and delivery accuracy.

Stakeholders consistently prioritized visualization simplicity, intuitive navigation, and real-time information accessibility over complex analytical features, reinforcing the importance of user-centered system development.

3.2 Prototype Development

Based on the user requirements identified during the Empathize and Define stages, an intelligent digital dashboard prototype was developed to provide comprehensive production visibility throughout the automotive aftermarket manufacturing process. The prototype was designed as a web-based application that could be integrated with the company's existing Enterprise Resource Planning (ERP) infrastructure without requiring substantial modifications to the current operational system.

Unlike conventional production monitoring applications that present isolated operational data, the proposed dashboard integrates production planning, inventory monitoring, work-in-process tracking, delivery scheduling, and managerial performance indicators into a unified visualization platform. The dashboard emphasizes simplicity, usability, and accessibility, allowing operational personnel and managers to obtain relevant information according to their organizational responsibilities.

The development process followed an iterative prototyping strategy. Initial interface sketches were first created based on user personas and problem statements derived from the Design Thinking process. These conceptual layouts were subsequently refined through collaborative discussions involving managers, production administrators, ICT personnel, purchasing staff, marketing representatives, and production operators. User feedback obtained during each iteration guided interface modifications, information grouping, navigation improvements, and visualization refinement before the prototype reached its final version.

The dashboard homepage provides a high-level overview of organizational performance by presenting key operational indicators such as active purchase orders, production progress, material availability, work-in-process status, delayed production activities, delivery performance, and key performance indicators (KPIs). Interactive charts, progress indicators, color-coded alerts, and filtering options enable users to identify operational problems quickly and prioritize corrective actions.

A major innovation introduced by the prototype is the implementation of barcode-supported production tracking. Each production batch is assigned a unique barcode that accompanies the product throughout every manufacturing stage, including material receiving, stamping, welding, Electronic Dip Painting (EDP), packaging, warehousing, and delivery. Barcode scanning automatically updates production status within the dashboard, eliminating manual recording activities while significantly improving production traceability.

Furthermore, the dashboard incorporates automated notification mechanisms that immediately inform users when predefined operational conditions occur. Examples include delayed production processes, material shortages, overdue purchase orders, inventory below minimum thresholds, and shipment delays. These alerts enable proactive managerial intervention before operational disruptions affect customer deliveries.

Overall, the prototype demonstrates how user-centered digital technologies can transform fragmented production information into an integrated decision support platform capable of supporting both operational execution and strategic management.

3.3 Intelligent Dashboard Architecture

The proposed intelligent dashboard adopts a multilayer architecture that integrates heterogeneous production information into a centralized visualization platform. The architecture consists of four primary layers: the data source layer, integration layer, application layer, and presentation layer.

The data source layer collects operational information generated from multiple organizational functions. These include purchase order information from ATPM, ERP databases, inventory records, production planning schedules, barcode scanning devices, warehouse transactions, quality inspection records, and delivery management systems. Since operational information originates from different departments, data standardization is necessary before visualization.

The integration layer functions as the core processing component responsible for consolidating production data from multiple sources. Data extraction, transformation, synchronization, and validation procedures ensure information consistency before being transmitted to dashboard modules. This architecture minimizes duplicate information while maintaining compatibility with the existing ERP system.

The application layer contains the operational modules developed during this research. These modules include Purchase Order Monitoring, Material Availability Management, Production Tracking, Barcode-Based Work-in-Process Monitoring, Delivery Scheduling, and KPI Analytics. Each module processes specific operational information while maintaining data consistency through centralized integration.

Finally, the presentation layer provides user-specific interfaces designed according to organizational responsibilities. Managers receive strategic performance summaries and KPI visualizations, whereas production operators access work-in-process tracking and barcode scanning interfaces. Purchasing personnel monitor inventory levels and procurement activities, while marketing

staff observe customer orders and delivery performance. This role-based information presentation prevents information overload while ensuring that every user receives relevant operational information.

The modular architecture also facilitates future system expansion. Additional intelligent functions such as production forecasting, predictive maintenance, machine learning-based scheduling, and artificial intelligence-supported decision recommendations can be incorporated without fundamentally modifying the dashboard structure.

3.4 Usability Testing Results

The proposed dashboard prototype was evaluated through usability testing involving representatives from all participating departments. The objective of the evaluation was to determine whether the developed dashboard effectively addressed user requirements identified during the Design Thinking process while supporting operational activities within the production environment.

Evaluation employed the Feedback Capture Grid method, enabling participants to categorize their experiences into four dimensions: positive aspects, opportunities for improvement, questions, and additional ideas. This qualitative evaluation approach provides richer insights than conventional questionnaire-based usability assessments because participants directly explain their operational experiences while interacting with the prototype.

Overall, users responded positively to the dashboard interface. Managers appreciated the availability of real-time production information presented through interactive visualizations that reduced dependency on manually compiled production reports. The centralized dashboard enabled rapid identification of delayed production activities, purchase order fulfillment status, and inventory conditions, thereby improving managerial responsiveness.

Production administrators reported that automatic synchronization between purchase orders, production schedules, and operational status substantially reduced repetitive administrative activities. The ability to monitor production progress without collecting reports from multiple departments was considered one of the dashboard's most valuable features.

Production operators indicated that barcode-based tracking simplified production recording and significantly reduced manual documentation errors. Instead of completing multiple paper-based forms, operators could simply scan product barcodes to update production status instantly. This functionality also enabled supervisors to identify product locations throughout the manufacturing process without physically visiting production stations.

Marketing personnel considered delivery monitoring particularly beneficial because shipment progress could be communicated more accurately to ATPM customers. Similarly, purchasing personnel appreciated real-time inventory visualization that facilitated procurement planning and reduced uncertainty regarding material availability.

ICT representatives highlighted the dashboard's compatibility with existing ERP infrastructure and emphasized that the proposed architecture minimizes implementation complexity while preserving organizational investment in current information systems.

Although the overall evaluation was highly positive, several participants suggested additional functionalities for future development. These included predictive production scheduling, supplier performance evaluation, mobile dashboard access, and artificial intelligence-based production recommendations. These suggestions demonstrate users' confidence in the proposed system while identifying opportunities for future enhancement.

3.5 Improving Production Visibility through an Intelligent Digital Dashboard

One of the principal findings of this study is that production inefficiencies in automotive aftermarket manufacturing are primarily associated with limited production visibility rather than insufficient manufacturing capacity. Although PT Adyawinsa Stamping Industries has implemented an Enterprise Resource Planning (ERP) system to support administrative activities, the production process remains highly dependent on fragmented information distributed across multiple organizational units. Consequently, managers experience significant delays in obtaining accurate production information, identifying operational bottlenecks, and making timely production decisions.

The intelligent digital dashboard proposed in this research addresses these limitations by integrating production planning, material availability, work-in-process tracking, delivery scheduling, and operational performance indicators into a centralized visualization platform. Unlike traditional reporting systems that generate static historical reports, the dashboard continuously updates operational information, allowing production managers to monitor manufacturing activities in real time. This capability substantially improves production transparency and reduces information latency across departments.

The findings support previous studies emphasizing the importance of production visibility within Industry 4.0 manufacturing environments. Real-time production information enables organizations to identify disruptions earlier, optimize production scheduling, and improve responsiveness to dynamic customer demand. However, unlike many previous dashboard implementations that primarily visualize existing ERP data, this study demonstrates that production visibility should be developed from a comprehensive understanding of user requirements rather than technological capability alone.

An important contribution of this research is the integration of operational information throughout the entire service part production lifecycle. Rather than monitoring individual manufacturing processes independently, the dashboard presents a unified operational perspective connecting customer demand, procurement activities, production execution, warehouse operations, and product delivery. Such holistic visualization enhances organizational awareness and facilitates collaborative decision-making among functional departments.

3.6 Design Thinking as a Human-Centered System Development Approach

The second important finding concerns the effectiveness of Design Thinking in developing intelligent production monitoring systems. Conventional manufacturing information systems are commonly designed from technical perspectives, emphasizing database architecture, system functionality, and software integration. Although technically sophisticated, these systems often fail to achieve high user acceptance because they inadequately address operational users' actual information requirements.

The Design Thinking methodology adopted in this research fundamentally shifted the system development process from technology-centered development toward user-centered innovation. Through iterative stages of Empathize, Define, Ideate, Prototype, Test, and Reflect, stakeholders from multiple departments actively participated in identifying operational challenges and validating proposed solutions. This participatory development process ensured that dashboard functionalities reflected real organizational needs rather than assumptions made by software developers.

The Empathize phase proved particularly valuable because operational challenges varied considerably among departments. Managers required strategic performance indicators, production administrators emphasized schedule synchronization, operators prioritized simplified production recording, purchasing personnel focused on inventory visibility, and marketing representatives required customer order monitoring. Understanding these diverse perspectives enabled researchers to develop an integrated dashboard that accommodates multiple organizational roles without overwhelming users with unnecessary information.

Furthermore, iterative prototype refinement significantly improved dashboard usability. User feedback collected during testing facilitated continuous interface improvements, resulting in intuitive navigation, simplified visualization, and enhanced information accessibility. These findings reinforce previous Design Thinking research suggesting that continuous user involvement substantially increases system usability and organizational adoption.

Therefore, this research demonstrates that Design Thinking is not merely a creative problem-solving methodology but also an effective framework for developing intelligent decision support systems within complex manufacturing environments.

3.7 Intelligent Dashboard as a Decision Support System

The intelligent dashboard developed in this research functions not only as a visualization platform but also as an operational Decision Support System (DSS). Traditional production reporting generally provides historical information after manufacturing activities have already occurred. Consequently, managerial decisions become reactive rather than proactive.

In contrast, the proposed dashboard continuously integrates operational information from multiple production activities, enabling managers to identify potential disruptions before they affect production performance. Early notification of material shortages, production delays, purchase order revisions, and delivery risks allows managers to implement corrective actions promptly.

Decision support is further strengthened through interactive visualization of key performance indicators, production progress, and inventory conditions. Rather than interpreting lengthy spreadsheet reports, decision makers can immediately identify operational priorities through graphical representations, color-coded alerts, and real-time performance summaries.

Another significant contribution is the implementation of barcode-based work-in-process monitoring. Automatic production status updates substantially reduce manual recording activities while simultaneously increasing production traceability. Accurate production information improves scheduling decisions, minimizes reporting errors, and strengthens coordination among departments.

The dashboard therefore supports tactical and operational decision-making simultaneously. Operational personnel receive detailed production information necessary for daily manufacturing activities, while managers obtain strategic performance summaries supporting production planning and resource allocation. This dual functionality distinguishes the proposed system from conventional production monitoring applications that primarily serve reporting purposes.



Fig 2. Dasbord System

The principal novelty of this research lies in the integration of Design Thinking, user-centered system development, and an intelligent digital dashboard into a unified decision support platform specifically designed for automotive aftermarket manufacturing. Unlike previous studies that focus primarily on ERP implementation or dashboard visualization, this research develops a human-centered production visibility system that integrates purchase order management, material availability, barcode-based work-in-process tracking, delivery monitoring, and KPI analytics within a single platform. This integrated approach contributes both theoretically to intelligent manufacturing research and practically to Industry 4.0 implementation by demonstrating how user-centered digital transformation can improve operational visibility, cross-functional collaboration, and manufacturing decision-making.

4. Conclusion

This study developed an intelligent digital dashboard using the Design Thinking approach to improve production visibility in automotive aftermarket manufacturing. The findings demonstrate that fragmented information management, manual production monitoring, and limited real-time visibility were the primary factors affecting operational efficiency and decision-making rather than production capacity itself. By systematically identifying user requirements through the Empathize, Define, Ideate, Prototype, Test, and Reflect stages, the proposed dashboard successfully addressed the operational needs of multiple stakeholders, including managers, production operators, administrators, purchasing personnel, marketing staff, and ICT specialists. The developed dashboard integrates purchase order monitoring, material availability, production tracking, barcode-based work-in-process monitoring, delivery scheduling, and key performance indicator (KPI) visualization into a single decision support platform. The evaluation results indicate that the proposed system enhances production transparency, facilitates cross-departmental collaboration, improves information accessibility, and supports faster and more accurate managerial decision-making. Furthermore, the implementation of user-centered design principles contributed to high usability and organizational acceptance, demonstrating that technological solutions should be developed based on users' operational requirements rather than solely on technical considerations. The principal contribution of this research lies in integrating Design Thinking with intelligent dashboard development to create a human-centered production visibility system specifically designed for automotive aftermarket manufacturing. Unlike conventional dashboard implementations that primarily focus on data visualization, the proposed system functions as an integrated decision support platform that strengthens communication, operational coordination, and real-time production management. Despite these contributions, the study was conducted within a single manufacturing company, which may limit the generalizability of the findings. Future research should validate the proposed framework across different manufacturing sectors and incorporate advanced technologies such as artificial intelligence, machine learning, predictive analytics, and Industrial Internet of Things (IIoT) integration. These enhancements would enable the dashboard to evolve from a production monitoring tool into an intelligent predictive decision support system capable of supporting autonomous and data-driven manufacturing operations in Industry 4.0 environments.

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Declarations

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Data and Software Availability Statements

The qualitative data collected during this study, including interview transcripts, observational notes, production workflow documentation, and prototype evaluation results, contain confidential

operational information belonging to PT Adyawinsa Stamping Industries. Therefore, these data are not publicly available due to company confidentiality and data protection policies.

The intelligent digital dashboard prototype developed in this research was created exclusively for research and evaluation purposes within the Service Part Division of PT Adyawinsa Stamping Industries. The prototype and supporting documentation may be made available by the corresponding author upon reasonable academic request and subject to approval from the company.

The data supporting the findings of this study are available from the corresponding author upon reasonable request, provided that such requests comply with the confidentiality agreement and data-sharing policies of PT Adyawinsa Stamping Industries

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